

Elicitation Techniques in Requirements Engineering

1. Interviews and Questionnaires

Description:

Interviews are one-on-one or small group sessions used to collect detailed information directly from stakeholders. Questionnaires are structured sets of written questions distributed to a larger group for responses.

When to Use:

- When in-depth insights are needed from key stakeholders.
- To collect structured feedback from a large audience quickly.

Pros:

- Allows clarification and follow-up questions (interviews).
- Scalable for many participants (questionnaires).
- Good for exploring both known and unknown requirements.

* Cons:

- Time-consuming (interviews).
- Poorly designed questions may lead to biased or unusable results.

Example:

- Interviewing a store manager to understand daily operations. ✓
- Sending a questionnaire to students about desired features in a campus mobile app.

System Context: University Registration System

The university plans to develop a new online system for student registration, course enrollment, and academic record tracking. The system must serve students, advisors, faculty, and administrative staff.

Stakeholders and Their Roles

Stakeholder

Role

- ✓ 1. Student End user who registers for courses, checks schedules, and views grades.
- ✓ 2. Academic Advisor Reviews student course selections, gives approval, tracks progress.
- ✓ 3. Lecturer Manages course capacity, prerequisites, and grading.
- ✓ 4. Registrar Staff Administers course offerings, enrollment limits, registration periods.
- ✓ 5. IT Administrator Maintains system infrastructure, security, and technical support.

A. Interview Questions (for in-depth, one-on-one or small group discussions)

Interview with Student

- Can you describe your current experience registering for classes each semester?
- What challenges do you face during registration (e.g., speed, course availability)?
- What features would you like to see in a new registration system?
- How do you prefer to receive updates—email, SMS, or app notification?
- Would you find it helpful if the system suggested courses based on your degree plan?

Interview with Academic Advisor

- How do you currently review and approve student course selections?
- What kind of data or reports would help you support students more effectively?
- Do you need alerts for students at risk of not graduating on time?
- Would automatic prerequisite checks improve your workflow?

Interview with Registrar Staff

- How do you currently manage course offerings and schedules?
- What are the key administrative tasks you perform during registration season?
- What reporting features would be most useful (e.g., enrollment stats, capacity)?
- Do you need override capabilities for special cases?

Interview with Lecturer

- How do you handle course capacity and prerequisites?
- Do you need to see student lists before the semester begins?
- Would you like the ability to send announcements or schedule changes through the system?

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B. Sample Questionnaire Items (for collecting data from a large group)

Use Likert scale (1 = Strongly disagree, 5 = Strongly agree)

Target: Students

1. I find the current registration system easy to use. ☒1 ☐2 ☒3 ☐4 ☒5
2. I can usually enroll in all required courses without difficulty. ☐1 ☐2 ☐3 ☐4 ☐5
3. I would like to receive real-time notifications for seat availability. ☐1 ☒2 ☐3 ☐4 ☐5
4. I want the system to show prerequisites before enrolling in a course. ☐1 ☒2 ☐3 ☐4 ☐5
5. I prefer an app-based registration system over a web-only system. ☐1 ☐2 ☐3 ☐4 ☐5

Target: Faculty / Advisors

1. The current system supports me in advising students effectively. ☐1 ☐2 ☐3 ☐4 ☐5
2. I would benefit from dashboards showing student progress. ☐1 ☐2 ☐3 ☐4 ☐5
3. I need custom filters to identify students at academic risk. ☐1 ☐2 ☐3 ☐4 ☐5
4. Course prerequisite checking should be automatic. ☐1 ☐2 ☐3 ☐4 ☐5
5. The ability to send bulk messages to advisees would be useful. ☐1 ☐2 ☐3 ☐4 ☐5

Tips for Using These Questions:

- Begin interviews with open-ended questions, then narrow down.
- Use questionnaires to validate or prioritize features discovered in interviews.
- Analyze the feedback to derive **functional requirements** (e.g., real-time alerts) and **non-functional requirements** (e.g., system responsiveness usability).

2. Workshops and Focus Groups

Description:

Workshops are structured sessions where multiple stakeholders collaborate in real-time. Focus groups gather a small group of users to discuss needs and opinions about the system.

When to Use:

- To align stakeholder viewpoints and resolve conflicting needs.
- To brainstorm new ideas and prioritize features collaboratively.

Pros:

- Encourages interaction and consensus.
- Efficient way to gather diverse perspectives.
- Builds shared understanding among stakeholders.

Cons:

- Needs skilled facilitation.
- Risk of dominant personalities influencing outcomes.

Example:

- A workshop with hospital staff to design an online patient scheduling system.
- A focus group of teachers to refine e-learning platform features.

3. Brainstorming Sessions

Description:

A creative, open-ended session where participants suggest as many ideas or requirements as possible in a short time.

When to Use:

- Early stages of a project to explore potential features or improvements.
- When innovation and new ideas are the goal.

Pros:

- Encourages creativity and inclusion of all voices.
- Fast and energetic.

Cons:

- May generate unrealistic ideas.
- Needs filtering and organization afterward.

Example:

- Brainstorming new features for a bookstore loyalty app with staff and customers.

4. Observational Methods (Job Shadowing / Ethnography)

Description:

The analyst observes users in their real working environment to understand how they perform tasks and use current systems.

When to Use:

- To uncover unspoken needs, pain points, and workarounds.
- When users find it hard to explain what they do.

Pros:

- Reveals real behaviors, not just what users say they do.

- Useful for process automation and UI design.

Cons:

- Time-intensive
- May affect user behavior (observer effect).

Example:

- Observing receptionists in a clinic to understand appointment handling.

5. Document Analysis

Description:

Reviewing existing documents such as reports, policies, forms, logs, manuals, or legacy system documentation to understand current processes and regulations.

When to Use:

- To establish baseline knowledge before meeting stakeholders.
- When access to users is limited or restricted.

Pros:

- Useful historical and technical insight.
- Helps discover regulatory or legal requirements.

Cons:

- Information may be outdated or incomplete.
- Interpretation without context may be misleading.

Example:

- Analyzing paper-based inventory records to develop a new digital inventory system.